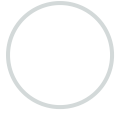


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Correction on HW 2 and lecture on DH parameters

김정현

All Sections

It turns out that the my lecture on DH parameters and coordinate assignment on articulated figures were rather quite incomplete.

1. The DH parameters I taught were based on "modified" DH parameters. Whether you use the original or modified version does not matter.

While the definition in the ppt is correct, to apply it correctly, the coordinate assignment (where to place x, y and z) is more complicated than I knew ... (it turns out I was taught with only simple examples ...). There is a more involved rules in assigning the coordinate system (not as simple as setting z to axis of rotation .. there are other things to consider ... if interested, see p. 79 of attached text). Based on those constraints, the full procedure to assign coordinate systems by DH convention is illustrated in p.82 of the attached text.

2. DH method actually simplifies the derivation of kinematics, but still one must understand the exact rules I mentioned above (e.g. p. 78). For use, we do not actually need to know this right now.

So here is what you should do.

1. Place x, y, z axis according to what I taught: z on the axis of rotation, x along the link (if possible, otherwise some other direction) and y will be determined automatically.

2. Then just figure out ${}^{i-1}R_i$ and ${}^{i-1}t_i$ (Once you have, ${}^{i-1}R_i$ and ${}^{i-1}t_i$, then you can compose ${}^{i-1}T_i$ easily) for the initial pose. For ${}^{i-1}R_i$, just express, axes of i with frame i-1 and lay them in the columns. For ${}^{i-1}t_i$, just figure out how much to translate to frame i with respect to i --> **T1**

2. Then just figure out, another the effect of rotation of joint i-1.

--> Typically, rotation of joint i-1 will be rotation of z_{i-1} . We know the formula for this.

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So the **T2** will be: like below with

$R_z(\theta)$	0
	0

			0
0	0	0	1

and the ${}^{i-1}\mathbf{T}_i = \mathbf{T}_2 * \mathbf{T}_1$

This process seems a bit complicated ... this is why DH parameters are used to simplify this very process. But maybe learning the details of DH method (for now) is more complicated because, for us, we deal with simple human models with not to many levels of articulation. In robotics, or more refined and complicated human model, the use of DH method would be beneficial.

BTW, if you don't understand the above process, send me email or visit me and I will be happy to go over it again.

In summary, no need to to use DH method (unless you really want to learn the details with the attached text). Method above should be simple enough.

GJK

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